

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer 2024 Course: B. Tech. Branch : Mechanical Engineering & Allied Semester : VI Subject Code & Name: BTMEC604B_Y19 IC Engines Max Marks: 60 Date: 21/06/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Define the Following terms with neat sketch : 1. Swept Volume 2. Clearance volume 3. Compression ratio	CO-2	6
B)	Classify the IC engine Based on i) Working Cycle ii) Number of Strokes iii) Method of Governing iv) Field of Application	CO-1	6
C)	Compare two stroke and four stroke engines in detail.	CO-1	6
Q.2	Solve Any Two of the following.		12
A)	Explain with the help of P-θ diagram, different stages of combustion in Spark Ignition engine.	CO-3	6
B)	Explain with labelled diagram normal and abnormal combustion in SI engine	CO-4	6
C)	Explain the factors influencing the flame speed with respect to i) Turbulence ii) Fuel- air ratio iii) Compression ratio	CO-4	6
Q. 3	Solve Any Two of the following.	CO-4	12
A)	Explain Magneto Ignition system of Engine with neat labeled sketch	CO-4	6
B)	State the different methods to measure frictional power of IC engines. Explain Morse test method in detail.	CO-3	6
C)	Compare Wet sump and Dry sump lubrication system.	CO-4	6
Q.4	Solve Any Two of the following.		12
A)	Explain in detail i) Centrifugal Supercharger ii) Vane type super charger with neat diagrams	CO-1	6
B)	State and explain BS-VI norms in detail for SI engines light and heavy motor vehicle	CO-4	6
C)	What is the effect of supercharging on the following parameters i) Power output ii) Mechanical efficiency iii) Fuel consumption	CO-1	6
Q. 5	Solve Any Two of the following.		12

A)	Explain alcohols as alternate fuel for IC engines with their merits and demerits	CO-3	6
B)	What are the types of fuel cell? Explain anyone.	CO-3	6
C)	Compare Electric vehicle with Hybrid Electric vehicle and list advantages of Hybrid vehicle	CO-4	6
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